

Ngoc (Ruby) Ngo

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EDUCATION

New York University

Master's in Data Science

New York, NY

Expected Graduation May 2027

Dickinson College

Bachelor of Arts in Quantitative Economics & Music-Performance Emphasis

Carlisle, PA

Sept. 2018 – May 2022

- GPA: 3.92, Honors: Summa Cum Laude, Phi Kappa Sigma, Dean's List

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, Splunk, Shell Scripting

Quantitative Skills: GenAI Ethics, Machine Learning, Econometrics, Bayesian Inference, Optimization, Statistical Modeling

Software: Kafka, PyCharm, Snowflake, Hadoop, Spark, RStudio, Tableau, WinSCP, Jira, Jenkins, React, Bitbucket, Git

WORK EXPERIENCE

Bank of America | Linux, Python, Splunk, Oracle SQL, Ansible, Jira

New York, NY

Software Engineer II

Feb. 2024 – Jul. 2025

- Saved **\$2.5M** in potential FINRA fines by building **Python automation**; repaired **100M+** malformed trading records
- Redesigned the repair execution workflow for flexibility and usability, cutting technical debt by **15%**
- Led upgrades and migrations of **RHEL-based systems**, modernizing legacy components for reliability and performance

Global Technology Analyst - Business Analyst

Jul. 2022 – Feb. 2024

- Analyzed **50M+** corrupt FIX messages post-trade to identify root causes and resolve systemic **data compliance** issues
- Developed **Python/Shell scripts** for **data reconciliation**; used comparative metrics to pinpoint discrepancies and cut manual correction volume by 40%

Genuine Parts Company | Stibo Systems, Python, Qlik, Snowflake

Atlanta, GA

Search Data Analyst Intern

May 2021 – Aug. 2021

- Created models to identify **1,000 products** with the highest discrepancy between physical and online sales, leading to a projected **20% increase** in online sales for these products with proposed strategies
- Improved efficiency of the data authoring assignment process by **50%** with automation by building a workflow in Stibo Systems' Workbench and screens in Web UI

Mekong Capital | SQL, Tableau, Trello

Remote

Vision Driven Data Analyst Intern

Jun. 2020 – Aug. 2020

- Identified trends and opportunities for negotiating down the **cost basis** of rental rates for commercial retail across the Mekong Capital portfolio by **30%**, based on pre- and current Covid-19 pandemic data analysis
- Generated reports utilizing **Tableau** on the productivity of trade areas in Ho Chi Minh City, identifying a **35% increase** in productivity in Districts 2 and 10, which led to reallocating restaurant resources in these areas

TECHNICAL PROJECTS

Copyright Laundering | OpenRouter, Python, MCP, OpenClaw, Docker

New York, NY

GenAI Accountability Data Scientist

April. 2026 – May. 2026

- Developed a novel **red-teaming framework** to evaluate copyright laundering in **autonomous agents**, identifying an awareness-action gap where agents leveraged **multi-step tool workflows** to bypass standard safety guardrails
- Engineered a benchmarking suite using **MCP** and **OpenClaw** to evaluate agentic tool selection across 10+ tasks, quantifying compositional risk in autonomous systems

Cartier Jewelry Recommender | Supabase, Python, MongoDB, MySQL, React, Vercel

New York, NY

Data Scientist

Oct. 2025 – Dec. 2025

- Built an **end-to-end** personal stylist app leveraging outfit images and **in-database cluster metadata** to recommend across 500+ jewelry items
- Designed a **Supabase PostgreSQL** backend with **Edge Functions**, logging recommendation sessions across 25+ relational tables with low-latency queries

Google AI Hackathon - SnapEat | Python, Google Gemini, Azure, Django, Figma

Remote

Back-end Engineer

Apr. 2024

- Developed a **Generative AI-powered menu recommendation system** to support decision-making under user constraints, validated through **100+** test inputs to ensure consistent performance
- Mapped out the product's **user journey** through a comprehensive flowchart, encompassing +10 key touchpoints within the app; this visualization helped identify potential **pain points** and opportunities for improvement

EXTRA-CURRICULAR & HONORS AND AWARDS

Bank of America's Asian Leadership Network, Data Analytics Lead 2024 & 2025

New York, NY

New York Piano Society's 2024 Carnegie Hall Performing Member

New York, NY

Women in Economics Club, Vice President of Programming 2021 & 2022

Carlisle, PA

Grace Hopper Celebration 2020 & 2021 & 2025 & 2026 Scholar, Two Sigma Tapia 2020 & 2021 Scholar

Remote